

ISyE 3030 Project: Technology's Impact on Quality of Life

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I. Introduction

Today, screen time has become a central part of daily life across all age groups. With the rapid increase of smartphones, computers, and other electronic devices, people spend more time in front of screens. This project aims to investigate how daily screen time influences various indicators of well-being, including mental health, stress levels, sleep duration, and physical activity. Additionally, by separating the data by age group, we can assess whether these effects differ significantly across age groups. By leveraging the publicly available dataset "Mental Health and Technology Usage" from Kaggle (*Waqar Ali, 2023*), we will analyze patterns in screen time and assess their relationship to well-being by conducting a comprehensive statistical analysis, which includes hypothesis testing, regression modeling, and application of advanced statistical techniques. The goal of this project is to determine if screen time is merely a byproduct of modern living or if it is a factor that significantly influences the quality of life, and if so how its impact varies across different age groups. This analysis could offer insights into healthier screen habits and inform future guidelines or interventions aimed at promoting mental and physical health in a technology-driven society.

II. Data Collection and Preprocessing

We downloaded the "mental_health_and_technology_usage_2024.csv" public dataset from Kaggle (*Waqar Ali, 2023*) as our collection method. The dataset contains self-reported information on individuals' screen time, technology usage habits, and well-being indicators such as sleep duration, stress levels, and mental health status. To process the data, we inspected its structure using pandas and numpy functions (`np.isnan()`, `df.dtypes`, `groupby...`) to check for missing values, verified data types, and ensured continuous variables like "ScreenTime" fall within acceptable ranges (0–24 hours). After filtering missing values by dropping rows with null entries, converting categorical variables like "AgeGroup" into factors, and validating data, we cleaned and formatted the dataset, consisting of 10,000 rows and 14 columns, and saved it to a new CSV file for further analysis.

III. Descriptive Statistics

To build a comprehensive understanding of behavioral and lifestyle patterns in the dataset, we started with a thorough numerical exploration. For the descriptive statistics of our dataset, we calculated the mean, median, mode, variance, quartiles, and skewness for twelve key behavioral and derived variables, including age, sleep hours, gaming hours, and total tech usage. Age was approximately normally distributed, with a mean of 41.52 and skewness near zero (0.005). In contrast, variables like Sleep Loss Ratio (skewness = 3.03) and Digital Engagement Ratio (skewness = 2.85) showed strong right skew, indicating long-tailed distributions driven by outliers. This is supported by their large variances (1.72 and 5.60 respectively) and wide interquartile ranges (1.61 for Digital Engagement Ratio), highlighting the need to account for individual variability in digital behavior when analyzing impacts on sleep and well-being.

We then coded visualizations using the matplotlib (*Matplotlib Developers, 2024*) library to better understand the data distribution. For visualization 1, *figure 1* presents a boxplot comparing the Sleep Loss Ratio (Sleep Deficit / Screen Time Hours) across three age groups: Young, Middle_Aged, and Senior. The boxplots show similar medians and interquartile ranges across all groups, suggesting that the effect of daily screen time on sleep loss may not significantly differ by age. A few outliers, particularly among the Middle_Aged and Senior groups, indicate that some individuals experience disproportionately high sleep deficits. The lack of significant variation in sleep loss ratio suggests that screen time's effect on sleep deficit doesn't vary by age.

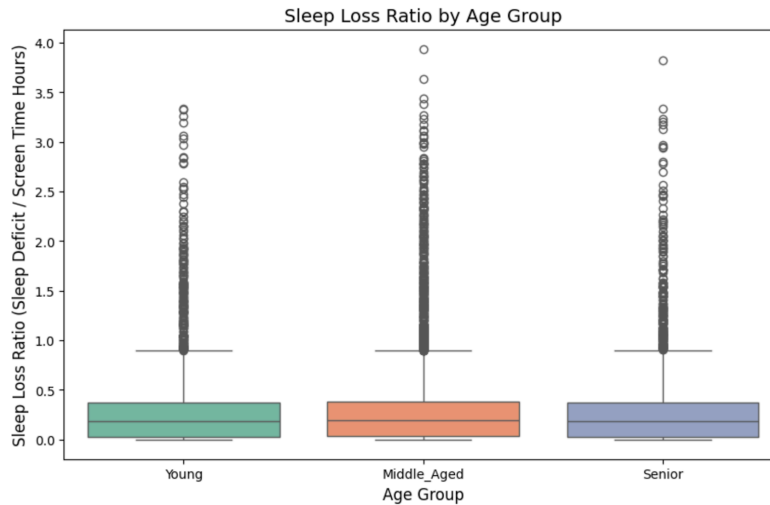


Figure 1- Boxplot of Sleep Loss Ratio by Age Group

For visualization 2, *figure 2* displays a scatter plot illustrating the relationship between Digital Engagement Ratio (Total Tech Usage / Screen Time Hours) and Sleep Loss Ratio (Sleep Deficit / Screen Time Hours). The regression line shows a positive relationship, suggesting that higher digital engagement may be associated with more sleep deficits.

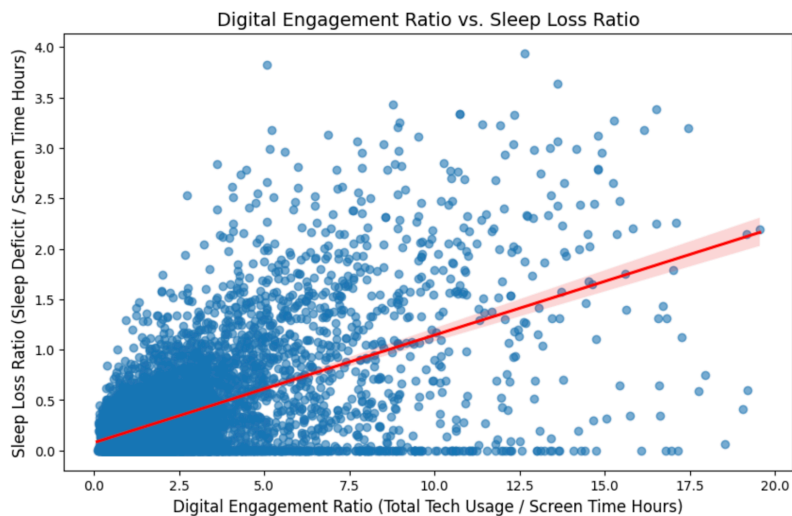


Figure 2- Scatter Plot with Regression Line Digital Engagement Ratio vs. Sleep Loss Ratio

For visualization 3, *figure 3* presents a grouped bar chart comparing the average digital usage ratios: Social Media Ratio and Technology Ratio, across three distinct age groups: Young, Middle_Aged, and Senior. Social Media Ratios remain below 1.0, indicating social media occupies a smaller, though still significant, portion of daily screen engagement. Surprisingly, Young and Senior individuals exhibit nearly identical usage ratios. This could reflect increased digital literacy among older adults or similar time-use behavior patterns among younger individuals and tech-savvy seniors. Meanwhile, Middle_Aged users

show slightly lower ratios in both categories, potentially due to more structured schedules such as work or family time limiting their discretionary tech use.

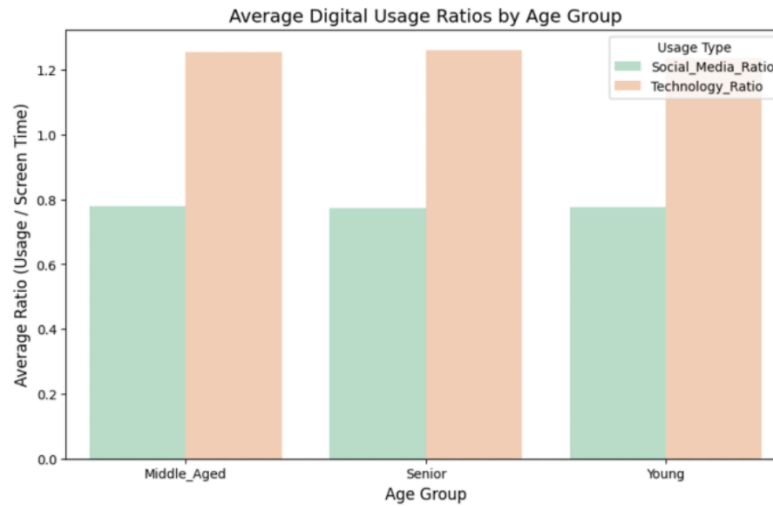


Figure 3- Grouped Bar Chart for the Average Digital Usage Ratios by Age Group

IV. Statistical Inference

Hypothesis: We tested whether the average sleep duration is significantly less than the medically recommended 8 hours.

$H_0: \mu = 8$ hours; $H_1: \mu < 8$ hours

Test Selection: A one-sided t-test is appropriate because the population standard deviation is unknown but can calculate the sample standard deviation, and there is a large sample size ($n = 10\,000$). We conducted the test at a 0.05 significance level.

Test Statistic Calculation:

Sample mean ($\bar{x}$) = 6.5 hours

Sample standard deviation = 1.45 hours

Sample size (n) = 10000

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

$$t = (6.5 - 8) / (1.45 / \sqrt{10,000})$$

$$t = -1.5 / 0.0145 = -103.45$$

Decision: With 9,999 degrees of freedom, the p-value is essentially zero ($p < 0.001$), which is less than our significance level of 0.05. We reject the null hypothesis.

Conclusion: We rejected the null hypothesis, meaning there is statistically significant evidence that the average amount of sleep is less than 8 hours. This supports concerns about widespread sleep deprivation in the context of high screen time usage.

V. Regression Analysis

We conducted a multiple linear regression to model the relationship between Sleep Loss Ratio (the ratio of sleep deficit to screen time) (dependent variable) and Digital Engagement Ratio, Screen Time Hours, and Age Group (independent variables). The model is:

$$\text{Sleep Loss Ratio} = \beta_0 + \beta_1 \cdot \text{Digital Engagement Ratio} + \beta_2 \cdot \text{Screen Time Hours} + \beta_3 \cdot \text{Age Group} + \epsilon_i$$

We used the Method of Least Squares to conduct our regression analysis:

- $R^2 = 0.349$: The model explains approximately 34.9% of the variance in Sleep Loss Ratio, indicating a moderate fit.
- F-statistic=1337.65, p-value= $2.33 \cdot 10^{-308} \approx 0.00$: The model is statistically significant, and the predictors together explain a significant portion of the variability in Sleep Loss Ratio.

Estimated coefficients and inference:

- Digital Engagement Ratio ($\beta_1 = 0.0817$, $p < 0.05$): Significant positive association with Sleep Loss Ratio. For each unit increase in Digital Engagement Ratio, Sleep Loss Ratio increases by 0.0817, when we hold other variables constant. This suggests that greater engagement with digital technologies is linked to higher sleep loss
- Screen Time Hours ($\beta_2 = -0.0212$, $p < 0.05$): Significant negative association, suggesting longer screen time may relate to less sleep loss. Interestingly, an increase in Screen Time Hours is associated with a decrease in Sleep Loss Ratio. For each additional hour of screen time, Sleep Loss Ratio decreases by 0.0212, which could suggest that longer screen time is linked to factors like sleep duration or timing that mitigate sleep loss.
- Age Group (Young and Senior): Both coefficients β_3 (-0.0080 and -0.0126 respectively) were negative, emphasising on a minor decrease in Sleep Loss Ratio but not statistically significant ($p > 0.05$).

Model adequacy was assessed by examining residual plots and a histogram of residuals. The histogram showed a roughly symmetric, bell-shaped distribution centered around zero, which supports the assumption of normally distributed residuals. This shape indicates that the model errors are not heavily skewed or biased.

Conclusion: Digital Engagement and Screen Time significantly impact Sleep Loss Ratio, while Age Group does not. These findings suggest technology behaviors, not age, are stronger predictors of sleep outcomes. And these results imply that interventions aimed at improving sleep quality should prioritize managing digital engagement behaviors rather than focusing only on screen time limits or age-specific strategies.

VI. Advanced Statistical Method

To explore the relationship between technology usage and sleep patterns, we applied a Neural Network modeling technique using Python, to predict Sleep Hours based on behavioral and demographic features. The model included predictors such as Age, Technology Usage Hours, Social Media Usage, Gaming Hours, Screen Time Hours, Stress Level, and Mental Health Status.

The model performed poorly, with an R^2 of -0.1122 and MSE of 2.35, indicating that it failed to capture the complex relationships in the data and that key influencing variables may be missing.

Despite this, feature importance analysis results (Figure 4) revealed:

- Gaming Hours had the strongest positive influence (~ 0.0125), followed by Screen Time Hours (~ 0.011) and Social Media Usage (~ 0.0085).
- Mental Health showed moderate importance, reinforcing known links between tech use and well-being.

- Technology Usage Hours and Age had lower but still notable effects (~ 0.005 each).
- Stress Level had a small negative importance (~ -0.0025), suggesting an inverse relationship with sleep.

These results highlight that while the model lacked predictive accuracy, it still offers insight into which behaviors most influence sleep outcomes.

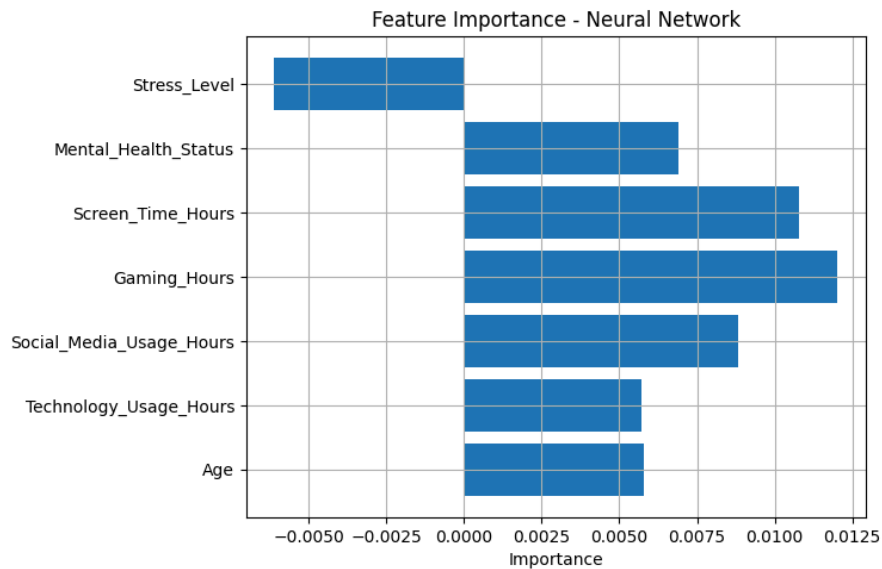


Figure 4- Permutation Feature Importance from Neural Network Model Predicting Sleep Hours

VII. Conclusion

This project explored the impact of daily screen time on well-being indicators such as sleep, stress, and mental health across different age groups. These analyses revealed that average sleep duration is significantly lower than the recommended 8 hours. Regression analysis showed that higher digital engagement is associated with increased sleep loss, while overall screen time showed a slight, inverse relationship. Surprisingly, age group was not a significant factor, suggesting these effects are consistent across generations. However, the study has limitations: it relies on self-reported, cross-sectional data, which limits causal interpretation and may include bias or inaccuracies. The underperformance of the neural network model (negative R^2) also suggests that key variables influencing well-being may be missing. Future research should include exploring more diverse behavioral and contextual variables and applying non-linear or personalized modeling approaches to better capture the complex relationships between screen time and well-being.

Citations

Waqar Ali (username: waqi786). *Mental Health & Technology Usage Dataset*. Kaggle, 2023, <https://www.kaggle.com/datasets/waqi786/mental-health-and-technology-usage-dataset>

Matplotlib Developers. *Plot Types — Matplotlib 3.8.2 Documentation*. 2024, https://matplotlib.org/stable/plot_types/index.html